

Sanando Banerjee

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PROFESSIONAL SUMMARY

Software Engineering undergraduate focused on backend and AI systems, with hands-on experience building API-driven, real-time, and AI-powered applications. Experienced in designing modular backend services, data pipelines, and full-stack systems using Python and modern web technologies.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript

Backend: FastAPI, REST APIs

Frontend: React, Next.js

Databases: MongoDB, SQL Server

AI/ML: LangChain, LangGraph, RAG, MCP

Tools & Testing: Git, Docker, Linux, Pytest, Playwright

PROJECT EXPERIENCE

Saturn-AI – Real-Time AI Trading Intelligence Platform

- **Engineered** a real-time Bitcoin intelligence platform that ingests news from **5 RSS sources every 10 minutes**, performs AI-powered sentiment analysis, aggregated into **BUY/SELL/HOLD signals with confidence scores**.
- **Designed** a modular FastAPI backend with **10 REST endpoints**, MongoDB persistence, scheduled processing via APScheduler, and a streaming AI explanation service for real-time market analysis.
- **Integrated** a React dashboard, CoinGecko price data, and an **MCP server** for AI-agent access; added **Playwright E2E tests** and **GitHub Actions CI/CD** for automated validation.

DDoS Overwatch – Real-Time Cyberattack Visualization Platform

- **Built** a real-time DDoS visualization platform that streams simulated attack telemetry and renders **animated source-to-destination attack flows** on an interactive world map.
- **Designed** an event-driven visualization pipeline where simulated attack events are streamed to the frontend and mapped to geospatial flows with visual encodings for attack intensity and provenance.
- **Developed** a responsive Next.js/TypeScript interface with **interactive map controls**, **real-time simulation controls**, and extensible hooks for region- and flow-level attack analysis.

RELEVANT EXPERIENCE

bITsera – Tech Club of IT, MSIT | Web Development Lead

Led and organized React workshops for 60+ students, covering state management, API integration and React hooks while emphasizing the importance of modular architecture and reusable component design.

EDUCATION

Meghnad Saha Institute of Technology

Bachelor of Technology (B. Tech) in Information Technology

Kolkata, India

Aug 2023 - Jul 2027
